

GymPAPPa

MANUAL TESTING REPORT

1. Manual Testing Overview

Manual testing was performed on the final running GymPAPPa system to verify key user and staff workflows through the browser interface. The tests focused on authentication, access control, equipment requests, court and crowd management, partner-finder workflows, role management, and equipment issue/return operations.

1.1 Test Environment

Application	GymPAPPa web application
Frontend	React / Vite browser interface
Backend	Node.js / Express
Database	PostgreSQL
Method	Functional manual testing using real application workflows
Evidence	Screenshots captured during execution

1.2 Result Summary

Test ID	Feature	Expected Outcome	Status
MT-01	Valid Student Login	Expected behavior observed	PASS
MT-02	Invalid Login Credentials	Expected behavior observed	PASS
MT-03	Protected Route / Unauthenticated Access	Expected behavior observed	PASS
MT-04	Equipment Request Submission	Expected behavior observed	PASS
MT-05	Equipment Request Validation	Expected behavior observed	PASS
MT-06	Court Status Update	Expected behavior observed	PASS
MT-07	Crowd Level Update	Expected behavior observed	PASS
MT-08	Partner Request Creation	Expected behavior observed	PASS
MT-09	Partner Join Request	Expected behavior observed	PASS
MT-10	Administrator Role Management	Expected behavior observed	PASS
MT-11	Counter Staff Equipment Issue	Expected behavior observed	PASS
MT-12	Counter Staff Equipment Return Flow	Expected behavior observed	PASS
MT-13	Administrator Event Creation	Expected behavior observed	PASS

Overall manual testing result: 13 / 13 documented scenarios passed.

2. Detailed Manual Test Records

MT-01 – Valid Student Login

Objective	Verify that a registered student can authenticate and access the student dashboard.
Precondition	GymPAPPa frontend and backend are running; a valid student account is available.
Expected Result	Authentication succeeds and the student dashboard is displayed.
Actual Result	The authenticated student dashboard loaded successfully and displayed the student's equipment availability view.
Status	PASS

Test Steps

1. Open the GymPAPPa login page.
2. Enter valid student credentials.
3. Select Login.
4. Confirm that the authenticated student dashboard is displayed.

The screenshot displays the GymPAPPa student dashboard. At the top, there is a header with the GymPAPPa logo and the text 'PERA Sports & Gymnasium Management System'. To the right of the header are 'Logout' and a user profile icon. Below the header is a navigation bar with links: Dashboard, Request Equipment, Partner Finder, Events & Tournaments, Court Availability, Request History, and Profile. The main content area features a welcome message 'Welcome, D.S. WELLAG' with a hand icon and a sub-message 'Track current equipment availability below and use the menu to view your request history.' Below this is a section titled 'Current Equipment Availability' with a search bar containing the text 'Search equipment — e.g. Racket, Shuttlecock' and a 'Clear' button. A table below the search bar lists equipment availability:

ID	Equipment	Sport	Available	Status
10	Badminton Racket	Badminton	12	Available
11	Shuttlecock	Badminton	20	Available

Figure 1 – Evidence for MT-01: Valid Student Login

MT-02 – Invalid Login Credentials

Objective	Verify that invalid or unusable login credentials are rejected with clear feedback.
Precondition	GymPAPPa login page is available.
Expected Result	Login is rejected and the user remains unauthenticated with an explanatory message.
Actual Result	The application rejected the login and displayed an explanatory account/password onboarding message.
Status	PASS

Test Steps

5. Open the login page.
6. Enter credentials that cannot complete password login.
7. Select Login.
8. Observe the validation/error feedback.

The screenshot displays the GymPAPPa login interface. At the top, it says "Welcome Back" and "GympAPPa - PERA Sports Management System". A red error message box states: "Password has not been set for this account. Please complete Google sign-in onboarding in your profile." Below this, the "User ID" field contains "e22402" and the "Password" field is filled with dots. A green "Login" button is present. Below the login fields, there is an "OR" separator and a "Sign in with Google" button. At the bottom, a link "Don't have an account? Register here" is shown, followed by the text "Use Google sign-in or your user ID with the password set in profile." On the right side, there are two informational sections: "About GympAPPa" which describes the app as a smart companion for sports and fitness at the University of Peradeniya, and "What you can do" which lists features like checking gym availability, borrowing equipment, finding teammates, joining tournaments, and following live scores. A third section, "Our Mission", states the goal is to make sports more accessible, organized, and enjoyable for every student at PERA.

Figure 2 – Evidence for MT-02: Invalid Login Credentials

MT-03 – Protected Route / Unauthenticated Access

Objective	Verify that unauthenticated users cannot remain on authenticated application pages.
Precondition	No active authenticated session.
Expected Result	The user is prevented from accessing the protected area and is returned to the login flow.
Actual Result	The login page was displayed while unauthenticated, preventing access to authenticated application functions.
Status	PASS

Test Steps

9. Log out or clear the authenticated session.
10. Attempt to access an authenticated area.
11. Observe the resulting page.

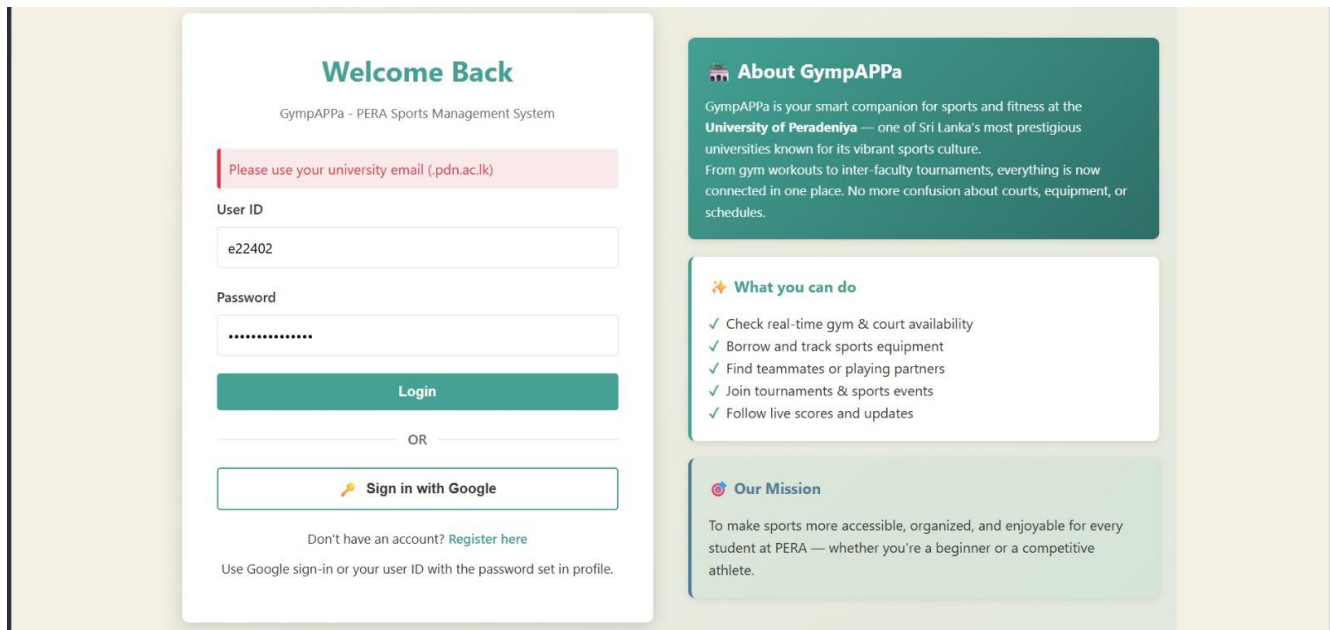


Figure 3 – Evidence for MT-03: Protected Route / Unauthenticated Access

MT-04 – Equipment Request Submission

Objective	Verify that a student can submit a valid equipment request.
Precondition	Student is authenticated and equipment is available.
Expected Result	The request is accepted and recorded as awaiting approval.
Actual Result	The application displayed “Pending — your request has been submitted and is awaiting approval.”
Status	PASS

Test Steps

12. Open Request Equipment.
13. Select a valid equipment quantity.
14. Select a valid pickup time.
15. Submit the request.
16. Observe the confirmation.

The screenshot shows the GymppAPa web application interface. At the top, there is a header with the PERA Sports & Gymnasium Management System logo and name, a 'Logout' button, and a user profile icon. Below the header is a navigation bar with links to Dashboard, Request Equipment (selected), Partner Finder, Events & Tournaments, Court Availability, Request History, and Profile.

The main content area displays the 'Badminton' section with a '2 items' indicator. It contains a table with the following data:

Equipment	Available	Request / Cancel	Requested by You
Badminton Racket	12	- +	0
Shuttlecock	20	- +	0

Below the table, there is a 'Select Pickup Time' section with a date and time picker showing 'mm/dd/yyyy --:-- --'. At the bottom, a green banner displays the message: 'Pending — your request has been submitted and is awaiting approval.'

Figure 4 – Evidence for MT-04: Equipment Request Submission

MT-05 – Equipment Request Validation

Objective	Verify that an incomplete equipment request is rejected instead of being submitted.
Precondition	Student is authenticated and the Request Equipment page is open.
Expected Result	The request is not submitted and a validation message is displayed.
Actual Result	The application blocked submission and displayed “No equipment selected to request.”
Status	PASS

Test Steps

17. Leave the equipment request quantity at zero/no item selected.
18. Choose a pickup time.
19. Attempt to submit the request.
20. Observe the validation message.

The screenshot shows the 'GympAPPa' interface for 'PERA Sports & Gymnasium Management System'. The user is on the 'Request Equipment' page for 'Badminton'. The page features a table with the following data:

Equipment	Available	Request / Cancel	Requested by You
Badminton Racket	12	- +	0
Shuttlecock	20	- +	0

Below the table, the 'Select Pickup Time' section shows a calendar selection for '09/26/2026 08:26 AM'. A green checkmark indicates 'Pickup set for: Sep 26, 2026, 8:26 AM'. At the bottom, a red error message states: 'No equipment selected to request.'

Figure 5 – Evidence for MT-05: Equipment Request Validation

MT-06 – Court Status Update

Objective	Verify that an authorized administrator can update a court status and receive confirmation.
Precondition	Administrator is authenticated and Court Management is available.
Expected Result	The new court status is saved and reflected in the management interface.
Actual Result	The court status update completed successfully and the interface displayed “Court status updated successfully.”
Status	PASS

Test Steps

21. Open Court Management.
22. Select a court.
23. Change its status.
24. Save the change.
25. Confirm the updated state/confirmation.

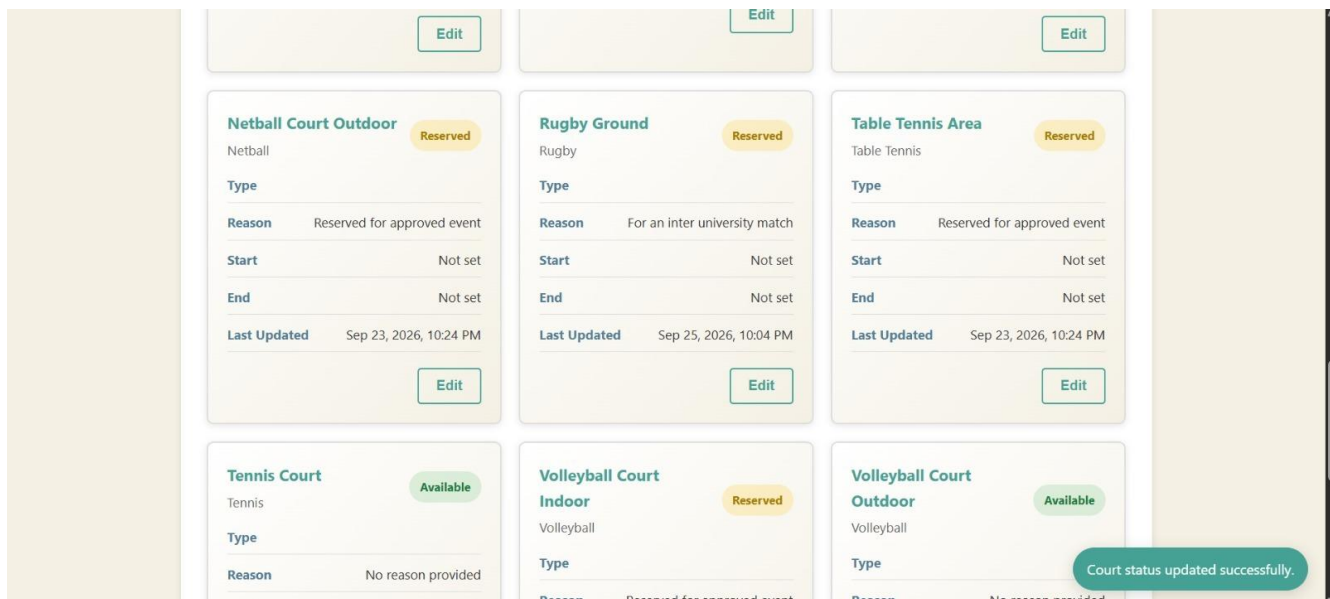


Figure 6 – Evidence for MT-06: Court Status Update

MT-07 – Crowd Level Update

Objective	Verify that authorized staff can update the current crowd level.
Precondition	Authorized staff member is authenticated.
Expected Result	The selected crowd level is saved and displayed as the current crowd level.
Actual Result	The crowd level was set to Moderate and the page displayed MODERATE as the current crowd level.
Status	PASS

Test Steps

26. Open Admin Court Management.
27. Select a crowd level.
28. Save the change.
29. Confirm the displayed crowd level.

The screenshot displays the GympAPPa web application interface. At the top, the logo and name 'GympAPPa' are visible, along with the tagline 'PERA Sports & Gymnasium Management System'. A navigation bar includes links for 'Events & Tournaments', 'Manage Equipment', 'Add Equipment', 'Court Management' (which is highlighted), 'Role Management', and 'Profile'. A 'Logout' button and a user profile icon are also present. The main content area is titled 'Admin Court Management' with a subtitle 'Monitor court availability, venue status, and gym crowd levels.' Below this, a 'CROWD MANAGEMENT' section is shown, featuring a 'Current Crowd' display with a 'MODERATE' button. A form below the display allows selecting a crowd level from a dropdown menu (currently set to 'Moderate') and includes a 'Save' button.

Figure 7 – Evidence for MT-07: Crowd Level Update

MT-08 – Partner Request Creation

Objective	Verify that a student can create a partner-finder request.
Precondition	Student is authenticated and Partner Finder is available.
Expected Result	A partner request is created and becomes available for matching.
Actual Result	The application displayed “Partner request created.” after submission.
Status	PASS

Test Steps

30. Open Partner Finder.
31. Enter the partner request details.
32. Select Create Request.
33. Observe the confirmation.

GympAPa
PERA Sports & Gymnasium Management System

Logout

Dashboard Request Equipment **Partner Finder** Events & Tournaments Court Availability Request History Profile

Create Partner Request D.S. WELLAGE

Sport: Badminton Date: mm/dd/yyyy

Start Time: --:-- -- End Time: --:-- --

8:00 AM – 8:00 PM only 8:00 AM – 8:00 PM only

Venue: Optional Skill Level: Intermediate

Gender Preference: Anyone

Additional Notes: Optional details

Create Request

Partner request created.

Figure 8 – Evidence for MT-08: Partner Request Creation

MT-09 – Partner Join Request

Objective	Verify that another student can request to join an open partner request.
Precondition	An open partner request exists and another student is authenticated.
Expected Result	A join request is sent for the selected open partner request.
Actual Result	The open partner request was displayed and the application confirmed “Join request sent.”
Status	PASS

Test Steps

34. Open Partner Finder.
35. Locate an open request.
36. Select Request to Join.
37. Observe the confirmation.

The screenshot shows the GympAPPa interface. At the top, there's a header with the logo, 'GympAPPa', and 'PERA Sports & Gymnasium Management System'. A 'Logout' button and a user profile icon are on the right. Below the header, a 'Create Request' button is visible. A red banner at the top of the main content area says 'Join request sent.' Below this, the 'Available Requests' section is displayed. It includes filters for Sport, Registration Number, Student Name, Date, Skill Level, and an Instant Search bar. A checkbox labeled 'Only Open Requests' is checked. The list shows one request for 'D.S. WELLAGE' with details: Reg no: e22425, Sport: Badminton, Date: 09/28/2026, Time: 4:15 PM, Skill: Intermediate, Venue: Any, Gender: Female, and 'No additional notes.' A green 'OPEN' tag is next to the request. A 'Request to Join' button is at the bottom of the request card. Below the card, another red banner says 'Join request sent.'

Figure 9 – Evidence for MT-09: Partner Join Request

MT-10 – Administrator Role Management

Objective	Verify that an administrator can update a registered user's role.
Precondition	Administrator is authenticated and Role Management is available.
Expected Result	The selected role is saved successfully.
Actual Result	The application displayed "User role updated successfully." after the role update.
Status	PASS

Test Steps

38. Open Role Management.
39. Locate a registered user.
40. Select the required role.
41. Save the role.
42. Observe the confirmation.

GympAPPa
PERA Sports & Gymnasium Management System

Logout

Events & Tournaments Manage Equipment Add Equipment Court Management **Role Management** Profile

Role Management

Filter users by registration number or user ID and update roles directly.

User role updated successfully

Registered Users & Direct Role Update

Search by registration number / user ID / name Clear

User ID	Name	Email	Current Role	New Role	Update
e22157	R.V. JAYALATH	e22157@eng.pdn.ac.lk	Student	Student	Save Role
e22425	D.S. WELLAGE	e22425@eng.pdn.ac.lk	Sports Counter Staff	Sports Counter Staff	Save Role
e22018	A.C.F. AKEELA	e22018@eng.pdn.ac.lk	Student	Student	Save Role
e22402	H.P.P.E. UMANDA	e22402@eng.pdn.ac.lk	Administrator	Administrator	Save Role
admin	Admin User	admin@pdn.ac.lk	Administrator	Administrator	Save Role

Figure 10 – Evidence for MT-10: Administrator Role Management

MT-11 – Counter Staff Equipment Issue

Objective	Verify that sports counter staff can issue approved equipment and update the request state.
Precondition	Counter staff is authenticated and a pending equipment request exists.
Expected Result	The requested equipment is issued and the request status changes accordingly.
Actual Result	The application confirmed that 2 Badminton Rackets were issued successfully and the record changed to Issued.
Status	PASS

Test Steps

43. Open Issue / Return Equipment.
44. Locate the student request.
45. Enter/confirm the issue quantity.
46. Select Issue.
47. Observe the confirmation and request status.

GympAPPa
PERA Sports & Gymnasium Management System

Logout

Equipment Stock Overview | **Issue / Return Equipment** | Issued Items History | Profile

Equipment Requests & Returns
Search by registration number fragment, then issue, decline, or return records with a quantity.

Search: e22425 Clear

✓ 2 Badminton Racket(s) issued to e22425 successfully!

ALL REQUEST RECORDS
Showing 6 of 11 records.

☒ All (11)
 ☐ Active (11)
 ☐ Completed (0)

#	Student ID	Equipment	Sport	Qty	Requested At	Pickup Time	Status	Issue / Return	Decline
1	e22425	Shuttlecock	Badminton	1	Sep 25, 2026, 10:29 PM	Sep 26, 2026, 10:29 AM	Pending Approval	1 Issue Decline	
2	e22425	Badminton Racket	Badminton	2	Sep 25, 2026, 10:29 PM	Sep 26, 2026, 10:29 AM	Pending Approval	2 Issue Decline	
3	e22425	Shuttlecock	Badminton	1	Sep 25, 2026, 08:23 PM	Sep 26, 2026, 09:00 AM	Pending Approval	1 Issue Decline	
4	e22425	Badminton Racket	Badminton	2	Sep 25, 2026, 08:23 PM	Sep 26, 2026, 09:00 AM	Pending Approval	2 Issue Decline	
5	e22425	Shuttlecock	Badminton	1	Sep 25, 2026, 08:23 PM	Sep 26, 2026, 09:00 AM	Pending Approval	1 Issue Decline	
6	e22425	Badminton Racket	Badminton	2	Sep 25, 2026, 08:23 PM	Sep 26, 2026, 09:00 AM	Issued	2 Return Decline	

Figure 11 – Evidence for MT-11: Counter Staff Equipment Issue

MT-12 – Counter Staff Equipment Return Flow

Objective	Verify that counter staff can initiate the return of previously issued equipment.
Precondition	Counter staff is authenticated and an issued equipment record exists.
Expected Result	The system presents a confirmation step before the return is finalized.
Actual Result	The return was confirmed successfully. The application displayed “1 Shuttlecock(s) returned successfully by e22425,” and the completed-record count increased.
Status	PASS

Test Steps

48. Open the issued equipment records.
49. Locate an issued item.
50. Select Return.
51. Verify that the system requests confirmation before completing the return.



GympAPPa
PERA Sports & Gymnasium Management System

[Logout](#)


Equipment Stock Overview

Issue / Return Equipment

Issued Items History

Profile

Equipment Requests & Returns

Search by registration number fragment, then issue, decline, or return records with a quantity.

Clear

✓ 1 Shuttlecock(s) returned successfully by e22425!

ALL REQUEST RECORDS

Showing 9 of 11 records.

All (11)
Active (9)
Completed (2)

#	Student ID	Equipment	Sport	Qty	Requested At	Pickup Time	Status	Issue / Return	Decline
1	e22157	Cricket Ball	Cricket	2	Jul 22, 2026, 03:51 PM	Jul 22, 2026, 06:51 PM	Pending Approval	2 Issue Decline	
2	e22157	Hockey Bat	Hockey	1	Jul 22, 2026, 03:50 PM	Jul 23, 2026, 03:50 PM	Pending Approval	1 Issue Decline	
3	e22157	Hockey Ball	Hockey	1	Jul 22, 2026, 03:50 PM	Jul 23, 2026, 03:50 PM	Pending Approval	1 Issue Decline	
4	e22157	Hockey Ball	Hockey	1	Jul 22, 2026, 03:49 PM	Jul 23, 2026, 03:49 PM	Pending Approval	1 Issue Decline	

Figure 12 – Evidence for MT-12: Counter Staff Equipment Return Flow

3. Manual Testing Outcome

The evidence-backed manual test run verified the principal end-user and staff workflows represented in this report. All 13 documented scenarios produced the expected observable behavior during testing.

Documented scenarios	13
Passed	13
Failed	0
Evidence	Screenshot evidence included for every documented scenario

3.1 Relationship to Automated Testing

This manual testing report complements the project’s automated testing. The automated suite verifies selected backend and frontend logic, while the manual tests validate browser-level workflows, user feedback, role-specific actions, and end-to-end interaction with the running system.

3.2 Defect and Validation Evidence

Manual testing confirmed that invalid or incomplete actions are blocked with user-facing feedback, including authentication feedback and equipment-request validation. The project testing documentation separately records the BE-001 equipment quantity validation defect and its automated regression verification.

3.3 Conclusion

The manual test evidence demonstrates that the tested GymPAPPa workflows operate as intended in the final application environment. The results, together with the automated test logs and coverage reports, provide verifiable testing evidence for the CO2060 final project evaluation.

MT-13 – Administrator Event Creation

Objective	Verify that an administrator can create a new event or tournament and that the created event is retained in the event list.
Precondition	Administrator is authenticated and the Events & Tournaments management page is available.
Expected Result	The event is created successfully and appears in the Created Events list with its approved status and scheduled date/time.
Actual Result	The application displayed “Event created successfully.” The created event was then visible in the Created Events section with an approved status and its scheduled date/time.
Status	PASS

Test Steps

52. Log in using an administrator account.
53. Open Events & Tournaments.
54. Enter valid event details and create the event.
55. Confirm that the success message is displayed.
56. Scroll to the Created Events section and verify that the newly created event is listed with its status and schedule.

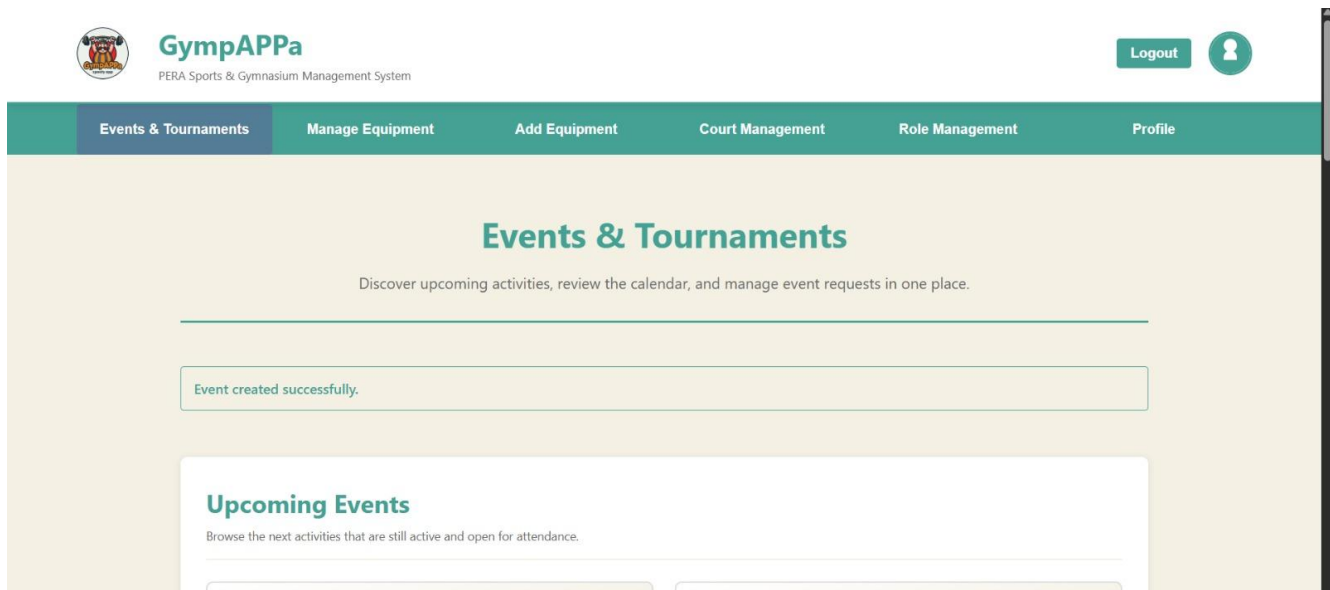


Figure 13.1 – MT-13 evidence: event creation completed successfully.

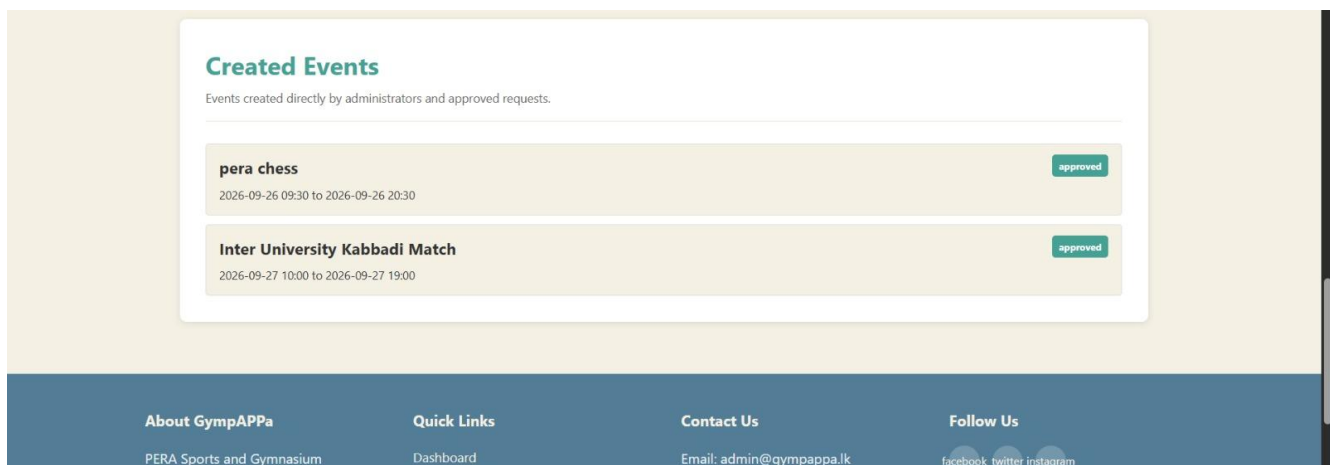


Figure 13.2 – MT-13 evidence: created event displayed in the Created Events list.

Final Submission Note

This report contains screenshot-backed manual testing evidence for 13 functional scenarios across student, administrator, and sports counter staff workflows. The manual evidence should be submitted together with the project's automated test results, coverage output, and defect-resolution record as the testing evidence for the CO2060 final evaluation.